Hybrid AI-based Detection of fileless malware in IOMT Healthcare systems

R26-CS-006

Project Proposal Report

Pagodaarachchi R. T. – IT22135070

B.Sc. (Hons) Degree in Information Technology Specialized in Cyber Security

Department of Computer Systems and Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

The rapid adoption of the Internet of Medical Things (IoMT) has significantly transformed modern healthcare by enabling connected medical devices, real-time patient monitoring, and data-driven medical services. However, this increased connectivity also introduces serious cybersecurity risks to healthcare infrastructures. IoMT environments are particularly vulnerable to sophisticated cyber threats such as ransomware, advanced persistent threats, and fileless malware that can compromise sensitive medical data and disrupt life-critical healthcare services. Existing cybersecurity mechanisms for healthcare networks primarily rely on reactive detection approaches, including intrusion detection systems and anomaly detection models that identify malicious activities only after they occur. This limitation highlights the need for proactive cybersecurity mechanisms capable of predicting potential threats before they impact healthcare systems.

This research addresses this gap by proposing a Predictive IoMT Threat Forecasting Engine designed to analyze historical IoMT network behavior and forecast potential cybersecurity threats in healthcare environments. The proposed system leverages machine learning–based predictive analytics to identify patterns in network traffic and detect early indicators of emerging cyber threats. By incorporating temporal analysis and predictive modeling, the framework aims to provide early threat intelligence that supports proactive defense strategies for healthcare infrastructures.

The system will utilize publicly available IoMT network datasets along with machine learning algorithms including Random Forest, Long Short-Term Memory (LSTM), and XGBoost for threat prediction and classification. Data preprocessing, feature engineering, and anomaly pattern analysis will be performed using Python-based data science libraries such as Scikit-learn, TensorFlow, and Pandas. The performance of the predictive models will be evaluated using standard metrics including accuracy, precision, recall, and F1-score.

The expected outcome of this research is the development of an intelligent threat forecasting framework capable of predicting potential cyber threats within IoMT healthcare networks. This approach aims to enhance the resilience of healthcare systems by enabling early threat detection and proactive cybersecurity responses. The proposed solution has potential social and commercial benefits by improving healthcare system reliability, protecting patient data privacy, and supporting the development of advanced cybersecurity solutions for healthcare organizations.

Keywords: IoMT Security, Predictive Cybersecurity, Machine Learning, Healthcare Cybersecurity, Threat Forecasting.

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TABLE OF CONTENT

DECLARATION ..... **Error! Bookmark not defined.**

ABSTRACT (Maximum 300 words) 4

ACKNOWLEDGEMENT 5

TABLE OF CONTENT 6

LIST OF TABLES 8

LIST OF FIGURES 9

LIST OF ABBREVIATIONS 10

LIST OF APPENDICES ..... **Error! Bookmark not defined.**

1. INTRODUCTION 11

1.1 Background And Context 11

1.2 Literature Review 12

1.3 Research Gap 13

2. OBJECTIVE 16

2.1 Main Objective 16

2.2 Specific Objectives 16

3. METHODOLOGY 16

3.1 Research Approach 17

3.2 Technical Methods And Algorithms 17

3.3 Tools And Platforms 18

3.4 Data Collection And Ethical Considerations 19

3.5 Validation Strategy And Performance Metrics 20

4. HIGH-LEVEL SYSTEM ARCHITECTURE 20

4.1 Overview Of The Proposed Solution And Its Components 20

4.2 High-Level System Diagram 22

4.3 Highlighting The Individual Component And Its Parts 22

4.4 Component Interactions 23

4.5 Data Flow Explanation 23

4.6 Scalability Considerations 24

4.7 Security Considerations 24

5. USER REQUIREMENTS 25

5.1 Stakeholder Identification 25

5.2 Functional Requirements 26

5.3 Non-Functional Requirements 27
COMMERCIALIZATION PLAN 28
6.1 Target Market and Customer Persona 28
6.2 Value Proposition 29
6.3 Revenue Model 29
6.4 Pricing Strategy 30
6.5 Competitive Advantage 30
6.6 Intellectual Property Considerations 31
6. WORK BREAKDOWN STRUCTURE (WBS) 32
7. GANTT CHART 33
8. REFERENCES LIST 34
9. APPENDICES ..... **Error! Bookmark not defined.**

LIST OF TABLES

Table 1: Comparison of Previous Research..... 15

LIST OF FIGURES

Figure 1.1: Hybrid AI-driven multi-layer security framework workflow diagram. 12
Figure 4.1: High-Level System Architecture of Component 3 – Predictive IoMT Threat Forecasting Engine 22
Figure 8.1: Work Breakdown Structure 32
Figure 9.2: Component 3 gantt chart. 33
Figure 9.1: Overall research project gantt chart 33

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DDoS	Distributed Denial of Service
DL	Deep Learning
IDS	Intrusion Detection System
IoMT	Internet of Medical Things
IoT	Internet of Things
IT	Information Technology
LSTM	Long Short-Term Memory
ML	Machine Learning
PACS	Picture Archiving and Communication System
RF	Random Forest
SDG	Sustainable Development Goal
SVM	Support Vector Machine
WMI	Windows Management Instrumentation
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

1.1 Background And Context

The rapid integration of the Internet of Medical Things (IoMT) into modern healthcare systems has significantly transformed medical services by enabling real-time patient monitoring, automated diagnostics, and efficient healthcare data management. IoMT consists of interconnected medical devices such as infusion pumps, wearable sensors, imaging systems, and remote monitoring tools that communicate through hospital networks and cloud infrastructure. While these technologies enhance healthcare efficiency and patient outcomes, they also expand the attack surface for cyber threats targeting critical medical infrastructure.

Healthcare organizations have increasingly become targets for cyberattacks due to the high value of medical data and the critical nature of healthcare services. According to the IBM Security Cost of a Data Breach Report, the healthcare sector has experienced the highest average breach cost among all industries, reaching USD 10.93 million per breach in 2023 [1]. Similarly, global reports indicate that ransomware and advanced persistent threats (APTs) targeting healthcare infrastructures have increased significantly over the past decade [2]. These attacks can lead to severe consequences such as disruption of medical services, manipulation of patient data, and compromise of life-critical medical devices.

IoMT environments are particularly vulnerable to advanced stealthy threats such as fileless malware, which operate entirely in system memory without leaving detectable files on disk. These attacks often exploit legitimate system tools such as PowerShell, Windows Management Instrumentation (WMI), and APIs, allowing attackers to evade traditional security mechanisms such as signature-based antivirus systems [3]. In healthcare environments, such attacks can propagate across networks, compromise imaging systems such as Picture Archiving and Communication Systems (PACS), and enable lateral movement between connected medical devices.

Existing cybersecurity solutions in healthcare environments primarily rely on signature-based detection systems, rule-based intrusion detection systems (IDS), and reactive monitoring tools. While these systems are capable of detecting known attack patterns, they often fail to identify unknown, evolving, or stealthy threats, particularly those associated with fileless malware attacks [4]. Furthermore, most existing solutions focus on real-time detection rather than predictive threat forecasting, limiting their ability to proactively mitigate threats before they escalate.

To address these challenges, this research proposes a Predictive IoMT Threat Forecasting Engine that utilizes machine learning techniques to analyze historical IoMT network behavior and security events to forecast potential cyber threats before they occur. The proposed system will integrate time-series forecasting models and anomaly detection techniques to identify emerging attack patterns and predict future threat activities within IoMT healthcare networks. By forecasting threats in advance, healthcare security teams can implement proactive defense strategies to reduce the risk of attacks on critical medical systems.

This research aligns with the United Nations Sustainable Development Goal (SDG) 3 – Good Health and Well-Being, which emphasizes the importance of secure and resilient healthcare systems. Ensuring the cybersecurity of IoMT infrastructures directly contributes to maintaining reliable healthcare services

and protecting sensitive patient data. Additionally, the research contributes to SDG 9 – Industry, Innovation and Infrastructure, which promotes the development of resilient technological infrastructure. From an academic perspective, this research aligns with the Cybersecurity and Artificial Intelligence research cluster, focusing on the application of AI-driven techniques to enhance security in critical digital infrastructures such as healthcare systems.

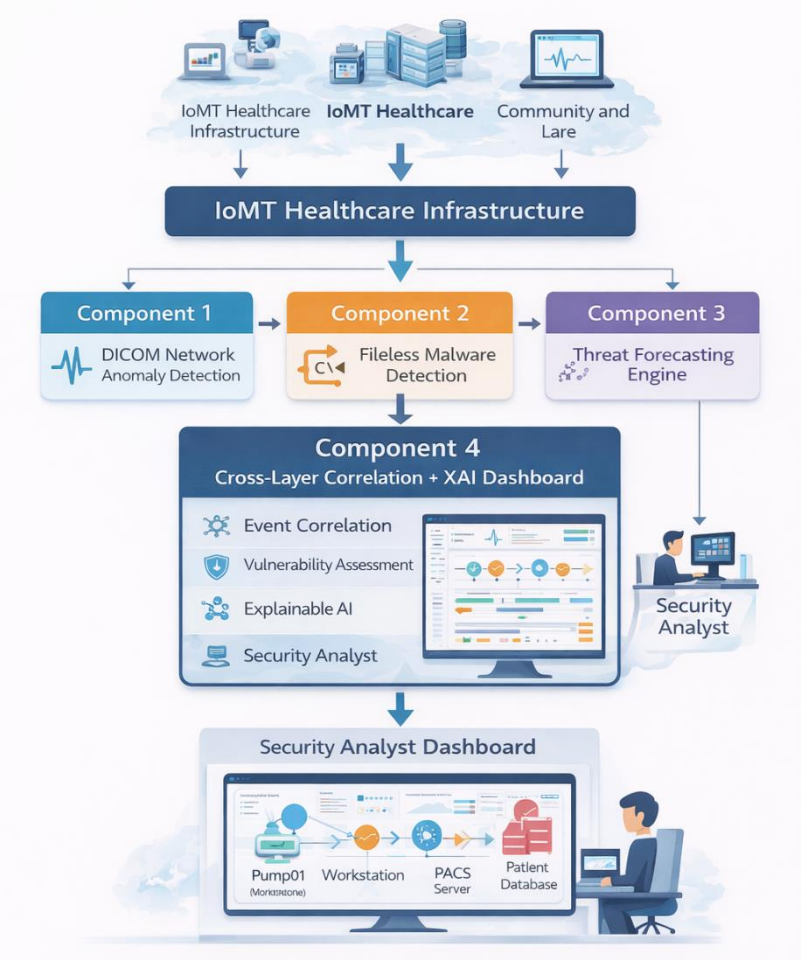


Figure 1.1: Hybrid AI-driven multi-layer security framework workflow diagram.

1.2 Literature Review

Recent research has explored various approaches to improving cybersecurity in IoMT environments using artificial intelligence and machine learning techniques. Several studies have focused on anomaly detection mechanisms for identifying abnormal network behavior within healthcare systems.

A study by Rahman et al. (2020) proposed a machine learning-based intrusion detection system designed specifically for IoMT environments, utilizing deep learning techniques to identify abnormal traffic patterns in healthcare networks [5]. Their research demonstrated that AI-based detection models could significantly improve detection accuracy compared to traditional rule-based systems.

Similarly, Alrashdi et al. (2021) introduced a deep learning framework for detecting cyberattacks in IoMT networks using recurrent neural networks (RNNs) to analyze sequential network traffic patterns

[6]. Their approach showed improved detection capabilities for previously unseen attacks, highlighting the potential of temporal learning models in healthcare cybersecurity.

Another study by Nguyen et al. (2022) examined the use of ensemble machine learning models for identifying malicious activity in IoMT devices. The researchers combined multiple machine learning algorithms to improve detection performance and reduce false positives in medical network environments [7].

In addition to academic research, several commercial cybersecurity solutions have been developed to protect IoMT infrastructures. Companies such as Medigate (now part of Claroty) and Armis Security provide healthcare-specific security platforms that monitor connected medical devices and identify suspicious network activity. These platforms use behavioral analysis to detect anomalies in medical device communications [8]. However, most existing products primarily focus on monitoring and detection rather than predictive threat forecasting.

Recent studies have also highlighted the importance of predictive analytics in cybersecurity. Buczak and Guven (2016) discussed how machine learning techniques can be used for predictive cyber threat intelligence by analyzing historical attack data to forecast potential threats [9]. Similarly, Sarker et al. (2021) emphasized the role of AI-driven predictive models in strengthening cybersecurity defenses by enabling proactive threat mitigation [10].

Despite these advancements, most existing research focuses on reactive intrusion detection mechanisms rather than predictive threat forecasting systems. Furthermore, many existing studies rely on general IoT datasets rather than datasets specifically designed for healthcare environments, limiting their effectiveness in real-world IoMT deployments.

1.3 Research Gap

Despite the increasing adoption of Internet of Medical Things (IoMT) technologies in healthcare environments, cybersecurity protection mechanisms for these systems remain limited and primarily reactive. Existing research and security tools focus largely on **detecting attacks after they occur**, rather than predicting potential threats before they emerge. This creates a significant gap in proactive cybersecurity protection for healthcare infrastructures.

Clearly Stating What Is Missing

Current IoMT security research mainly focuses on intrusion detection systems (IDS), anomaly detection frameworks, and rule-based monitoring mechanisms that identify malicious activities only after suspicious behavior has already appeared in the network. While these systems can identify abnormal network traffic patterns, they typically lack the ability to forecast future cyber threats based on historical attack trends and behavioral patterns.

Furthermore, many existing healthcare cybersecurity frameworks concentrate on device-level monitoring or network anomaly detection, without incorporating predictive analytics that could provide early warnings of emerging threats. As a result, healthcare organizations often remain vulnerable to

sophisticated cyberattacks such as ransomware campaigns, advanced persistent threats, and fileless malware that evolve over time.

Research such as the study by Alsubaei, Abuhussein, and Shiva (2019) highlights that IoMT environments face unique security challenges due to device heterogeneity, legacy medical systems, and limited built-in security mechanisms [1]. These characteristics make traditional detection-based security solutions insufficient for protecting critical healthcare infrastructures.

Why Current Approaches Are Insufficient

Many existing cybersecurity approaches rely on signature-based detection methods or conventional machine learning classification models that identify known attack patterns. These techniques struggle to detect zero-day attacks or evolving malware techniques, especially in IoMT environments where device communication patterns are highly dynamic.

Recent research on healthcare network security has explored machine learning approaches for detecting anomalies in medical systems. For example, the study "Analysis of ML Models in Detecting and Mitigating Anomalies in Healthcare Networks" demonstrates that machine learning models such as Random Forest, Support Vector Machines, and neural networks can improve anomaly detection accuracy in healthcare network environments [2]. However, the focus of such work remains on identifying anomalies in existing network traffic rather than forecasting future threat behavior.

Similarly, research in IoT security indicates that most machine learning-based solutions concentrate on classification tasks, where models categorize traffic as either benign or malicious. While effective for detection, these approaches do not analyze temporal patterns of attack evolution, which are necessary for predicting upcoming cybersecurity threats.

Another limitation is that many current studies rely on general IoT datasets, which do not accurately represent the specialized communication patterns of IoMT devices used in healthcare systems. This limits the effectiveness of existing security models when applied to real-world hospital networks.

Justifying Novelty and Creativity

The proposed Predictive IoMT Threat Forecasting Engine introduces a novel approach by shifting from traditional reactive detection mechanisms to proactive threat forecasting. Instead of only identifying attacks after they occur, the proposed system will analyze historical IoMT network data to identify temporal trends and behavioral indicators associated with cyber threats.

By integrating machine learning-based predictive models with time-series analysis techniques, the proposed solution aims to forecast potential cyber threats before they manifest in the healthcare network. This predictive capability allows cybersecurity teams to implement preventive measures and strengthen system defenses in advance.

Additionally, the proposed framework specifically targets IoMT healthcare environments, addressing the unique communication patterns and security challenges associated with connected medical devices. This domain-specific focus improves the relevance and applicability of the solution compared to general IoT security models.

Evidence-Based Argument

Recent studies emphasize the growing need for proactive cybersecurity strategies in healthcare infrastructures. According to Sarker et al. (2021), predictive analytics and machine learning techniques have significant potential to enhance cybersecurity defenses by enabling early detection and forecasting of cyber threats [3]. Similarly, surveys of IoMT security highlight the need for advanced AI-driven security mechanisms capable of adapting to evolving attack patterns in healthcare networks [1].

However, despite the recognized importance of predictive cybersecurity, relatively few studies have explored machine learning-based threat forecasting specifically for IoMT environments. Most research continues to focus on anomaly detection and post-incident analysis rather than predictive intelligence.

Therefore, this research addresses an important gap by proposing a predictive threat forecasting framework tailored to IoMT healthcare networks, combining machine learning models with historical network analysis to anticipate cyber threats before they occur. This contribution not only advances research in healthcare cybersecurity but also provides practical benefits for improving the resilience of medical infrastructures.

No	Feature	Research Paper 01 [1]	Research Paper 02 [2]	Research Paper 03 [3]	Proposed Component
1	Machine Learning for Threat Detection	✗	✓	✓	✓
2	Focus on IoMT Healthcare Networks	✓	✓	✗	✓
3	Predictive Threat Forecasting	✗	✗	✓	✓
4	Detection of Unknown Threats	✗	✓	✓	✓
5	Early Threat Warning for Healthcare Systems	✗	✗	✗	✓

Table 1: Comparison of Previous Research

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a machine learning–based Predictive IoMT Threat Forecasting Engine capable of analyzing historical IoMT network traffic and security events to forecast potential cyber threats in healthcare environments with improved early detection capability and predictive accuracy.

2.2 Specific Objectives

- To collect and preprocess IoMT network traffic and security event data from publicly available healthcare cybersecurity datasets in order to construct a structured dataset suitable for predictive threat analysis.
- To analyze historical IoMT network behavior and identify key threat indicators such as abnormal traffic patterns, suspicious communication behaviors, and attack-related anomalies within healthcare networks.
- To develop and train machine learning–based predictive models (e.g., time-series forecasting and anomaly detection models) capable of forecasting potential cyber threats in IoMT environments.
- To design and implement a Predictive IoMT Threat Forecasting Engine architecture that integrates data preprocessing, feature extraction, machine learning prediction models, and threat visualization components.
- To evaluate the performance of the proposed forecasting system using standard machine learning evaluation metrics such as accuracy, precision, recall, F1-score, and prediction reliability.
- To demonstrate how predictive threat forecasting can enhance proactive cybersecurity strategies in healthcare IoMT environments by enabling early threat mitigation and improving the resilience of medical systems against cyberattacks.

3. METHODOLOGY

This section describes the research approach, technical implementation methods, data collection process, and evaluation strategy used to develop the Predictive IoMT Threat Forecasting Engine. The methodology focuses on applying machine learning and predictive analytics techniques to identify and forecast cyber threats within Internet of Medical Things (IoMT) healthcare environments.

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach combined with an experimental evaluation method. Design Science Research is widely used in cybersecurity and information systems research for developing and evaluating innovative technological artifacts that address real-world problems.

The research aims to design and implement a Predictive IoMT Threat Forecasting Engine, which acts as the primary artifact of this study. The system will be developed to analyze IoMT network traffic patterns and predict potential cybersecurity threats before they occur.

The design science process in this research includes the following stages:

Problem Identification – Identifying cybersecurity challenges in IoMT healthcare environments, particularly the lack of predictive threat forecasting systems.

Objective Definition – Defining the goal of developing an AI-driven forecasting engine capable of predicting potential cyber threats.

Design and Development – Designing the architecture of the forecasting engine and implementing machine learning models.

Demonstration – Applying the developed system to IoMT network datasets to demonstrate its predictive capabilities.

Evaluation – Measuring system performance using standard machine learning evaluation metrics.

Communication – Documenting and presenting the research findings and system contributions.

This approach enables the development of a practical cybersecurity solution while simultaneously contributing to academic knowledge in IoMT security research.

3.2 Technical Methods And Algorithms

The proposed Predictive IoMT Threat Forecasting Engine consists of several technical components that process network data and generate predictive threat insights.

3.2.1 Data Preprocessing and Feature Engineering

The first stage involves preparing IoMT network traffic data for analysis. Raw dataset logs are processed to extract meaningful features such as: Network flow statistics, Packet size distributions, Communication frequency patterns, Protocol usage behavior, Source and destination IP characteristics, Time-based traffic patterns

Feature engineering techniques will be applied to transform raw network traffic into structured datasets suitable for machine learning models.

3.2.2 Threat Pattern Analysis

Exploratory data analysis (EDA) techniques will be used to identify patterns related to malicious activities. This includes analyzing: Anomalous communication patterns, Abnormal traffic spikes, Suspicious device behavior, Temporal attack patterns

These insights help identify features that are most relevant for predictive threat modeling.

3.2.3 Predictive Threat Forecasting Models

Machine learning models will be developed to forecast potential cybersecurity threats in IoMT environments. The research will evaluate different models including:

Random Forest Classifier
Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting. It is widely used in cybersecurity applications due to its robustness in handling complex datasets.

Long Short-Term Memory (LSTM) Networks
LSTM is a deep learning model designed for sequential and time-series data analysis. Since cyberattacks often exhibit temporal patterns, LSTM models can analyze historical network traffic to forecast potential future threats.

XGBoost (Extreme Gradient Boosting)
XGBoost is a highly efficient gradient boosting algorithm that provides strong performance for classification and predictive tasks. It is suitable for identifying complex relationships between network features and malicious activities.

The performance of these models will be compared to determine the most effective algorithm for predictive threat forecasting in IoMT environments.

3.3 Tools And Platforms

Several software tools and platforms will be used to implement the proposed system.

- **Python Programming Language**

Python will be used as the primary programming language due to its extensive ecosystem of machine learning and data analysis libraries.

- **Machine Learning Libraries**

The following libraries will be utilized for model development and evaluation:

Scikit-learn – For implementing traditional machine learning algorithms such as Random Forest and XGBoost.
TensorFlow / Keras – For developing deep learning models such as LSTM networks.
Pandas and NumPy – For data preprocessing and numerical computation.

- **Data Visualization Tools**

Visualization tools such as Matplotlib and Seaborn will be used to analyze network behavior and illustrate threat patterns.

- **Development Environment**

The system will be developed using Jupyter Notebook and Visual Studio Code, which provide an interactive environment for machine learning development and experimentation.

These tools were selected due to their strong support for data science workflows and their widespread adoption in cybersecurity research.

3.4 Data Collection And Ethical Considerations

The research will utilize publicly available IoMT cybersecurity datasets to ensure ethical compliance and data privacy protection.

The primary dataset considered for this research contains IoMT network traffic data representing both benign and malicious activities within healthcare environments. The dataset includes various attack scenarios that simulate real-world cybersecurity threats targeting medical devices.

Using publicly available datasets ensures that the research does not involve the collection of sensitive patient information or personally identifiable data.

If additional data collection is required, the research team will seek formal permission from healthcare institutions before accessing any hospital network data. All collected data will be anonymized and used strictly for academic research purposes.

Ethical considerations in this research include:

- Ensuring patient data privacy and confidentiality
- Avoiding the collection of personally identifiable information (PII)
- Using datasets only for legitimate academic research
- Following institutional ethical research guidelines

These measures ensure that the research complies with ethical standards for cybersecurity and healthcare data research.

3.5 Validation Strategy And Performance Metrics

The effectiveness of the proposed Predictive IoMT Threat Forecasting Engine will be evaluated using a structured validation process.

The dataset will be divided into training and testing subsets, typically using an 80:20 split, where the majority of the data is used for model training and the remaining portion is used for evaluation.

To ensure model reliability, k-fold cross-validation may also be applied to assess the consistency of model performance across different data partitions.

The following evaluation metrics will be used to measure the performance of the predictive models:

Accuracy - Accuracy measures the proportion of correctly predicted instances compared to the total number of predictions.

Precision - Precision evaluates how many of the predicted threats are actually malicious activities.

Recall (Detection Rate) - Recall measures the ability of the system to correctly identify actual cyber threats.

F1-Score - The F1-score combines precision and recall into a single metric that provides a balanced evaluation of model performance.

Confusion Matrix - A confusion matrix will be used to visualize the classification performance of the model by comparing predicted and actual results.

These metrics provide a comprehensive evaluation of the system's ability to accurately forecast potential cyber threats within IoMT healthcare environments.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

This section presents the overall architecture of the proposed solution and explains how the Predictive IoMT Threat Forecasting Engine operates within the wider healthcare cybersecurity environment. The architecture is designed to support proactive threat analysis by collecting IoMT-related data, processing it, identifying patterns, and generating future threat forecasts for healthcare networks.

4.1 Overview Of The Proposed Solution And Its Components

The proposed solution is an AI-driven Predictive IoMT Threat Forecasting Engine developed to analyze healthcare IoMT traffic and security events in order to forecast possible cyber threats before they escalate into attacks. The system is intended to function as a decision-support and early warning mechanism for hospital cybersecurity teams.

At a high level, the solution consists of the following main components:

IoMT Data Sources Layer
This layer contains the sources of data used by the system, such as IoMT network traffic datasets, medical device communication logs, and security event records. These sources represent both normal and malicious activities in healthcare environments.

Data Collection and Ingestion Layer
This component gathers raw traffic and log data from the selected datasets or healthcare simulation environments and prepares them for processing.

Data Preprocessing and Feature Engineering Layer
In this stage, raw data is cleaned, normalized, transformed, and converted into structured features suitable for machine learning analysis. Important features such as packet behavior, communication frequency, attack labels, and time-related patterns are extracted.

Threat Analysis and Pattern Learning Layer
This module examines historical IoMT data to identify suspicious behavior trends, attack signatures, anomalous communication patterns, and temporal threat indicators.

Predictive IoMT Threat Forecasting Engine
This is the main individual component of this research. It uses machine learning and time-series forecasting methods to predict the probability of future cyber threats based on historical patterns and current observations.

Alerting and Visualization Layer
The predicted threat results are displayed through dashboards, reports, and warning indicators so that administrators can understand the severity and likelihood of future attacks.

Security Response Support Layer
This final layer supports decision-making by allowing healthcare IT teams to use the generated predictions for proactive defense planning, monitoring, and mitigation.

4.2 High-Level System Diagram

Below is a suitable high-level architecture diagram for the proposal.

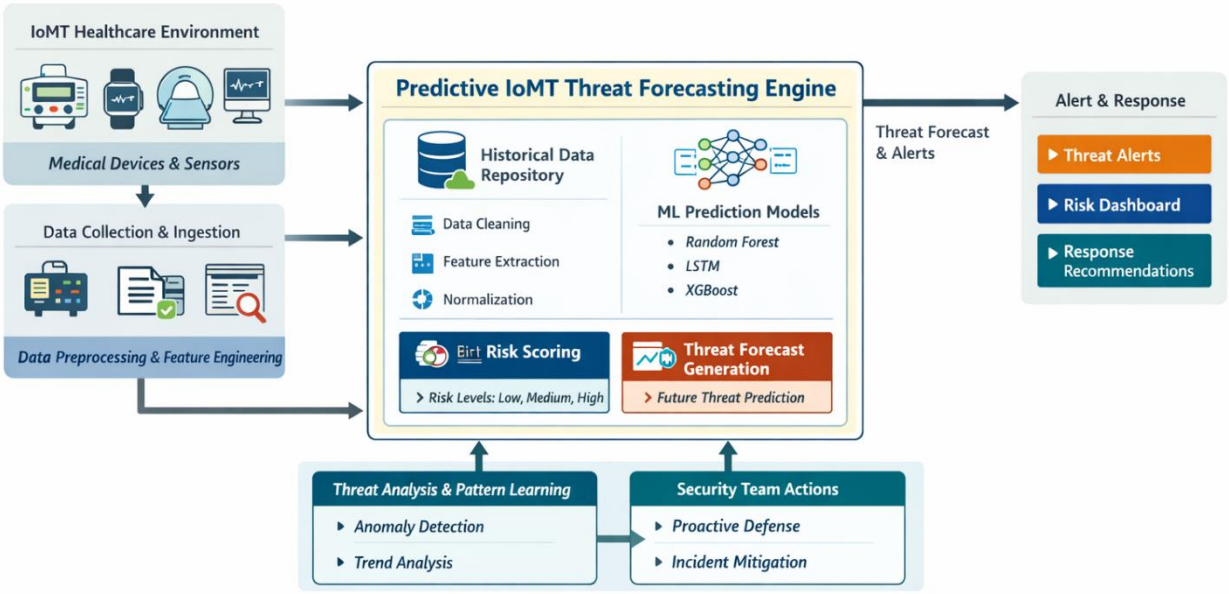


Figure 4.1: High-Level System Architecture of Component 3 – Predictive IoMT Threat Forecasting Engine

4.3 Highlighting The Individual Component And Its Parts

The main focus of this research is the **Predictive IoMT Threat Forecasting Engine**, which is the core intelligent component in the architecture. Its purpose is not only to detect suspicious activity but also to **forecast future threats** using learned attack behavior patterns.

The internal parts of this component include:

4.3.1 Historical Threat Data Repository

This sub-module stores processed historical IoMT traffic records, attack instances, anomaly scores, and other relevant security events. It acts as the knowledge base for model training and forecasting.

4.3.2 Feature Selection and Temporal Analysis Module

This module identifies the most relevant features for prediction, especially those related to time-dependent attack behavior such as burst traffic, repeated malicious communication, and abnormal protocol usage.

4.3.3 Machine Learning Prediction Module

This is the computational core of the engine. It applies selected machine learning algorithms such as:

- Random Forest
- XGBoost

- LSTM or other sequence-based forecasting models

These models learn patterns from historical IoMT security data and estimate the likelihood of upcoming threats.

4.3.4 Threat Risk Scoring Module

This part converts prediction outputs into interpretable risk levels such as **low**, **medium**, or **high** threat severity. This helps administrators understand the urgency of the forecasted threat.

4.3.5 Forecast Output and Alert Generation Module

This sub-module generates threat forecasts and transmits them to the dashboard or warning system. It provides the final predictive result of the component.

4.4 Component Interactions

The components of the proposed architecture interact in a sequential and logical manner.

First, the IoMT healthcare environment generates device communication data, security logs, and traffic records. These are collected through the data ingestion layer, which transfers them into the system.

Next, the preprocessing layer transforms raw records into structured and analyzable feature sets. These processed data points are forwarded to the threat analysis layer, where hidden attack trends and abnormal communication patterns are identified.

The output of the threat analysis layer is then passed into the Predictive IoMT Threat Forecasting Engine, which uses machine learning models to estimate the probability of future threats.

Finally, forecasted results are sent to the alerting and visualization layer, where the information is displayed to security administrators. Based on these outputs, the security response support layer assists in taking preventive actions before the predicted threat causes harm.

This interaction flow ensures that the system supports proactive rather than purely reactive cybersecurity operations.

4.5 Data Flow Explanation

The data flow of the proposed system can be explained in the following stages:

Step 1: Data Acquisition
Raw IoMT traffic data and security logs are obtained from datasets or simulated healthcare network environments.

Step 2: Data Ingestion
Collected data enters the system through the ingestion layer, where it is organized for further analysis.

Step 3: Data Preprocessing

The raw data is cleaned to remove missing, duplicated, or noisy records. Features are then extracted and normalized for machine learning processing.

Step 4: Pattern Discovery
The system analyzes the processed data to identify suspicious traffic characteristics, attack-related anomalies, and time-based patterns.

Step 5: Threat Forecasting
The predictive engine uses learned patterns and temporal behavior to forecast whether future network states are likely to be associated with a cyber threat.

Step 6: Threat Scoring and Output Generation
The prediction results are converted into risk levels and meaningful outputs such as forecast graphs, alerts, and reports.

Step 7: Administrator Decision Support
The final output is used by healthcare cybersecurity teams for early monitoring, preventive response planning, and improved incident readiness.

4.6 Scalability Considerations

The proposed architecture is designed with scalability in mind so that it can support growing IoMT healthcare environments.

First, the modular structure allows each layer to be improved or expanded independently. For example, additional IoMT device types or new traffic sources can be added to the data ingestion layer without changing the forecasting engine.

Second, the system can be extended to support larger datasets and real-time streaming in future implementations. If deployed in large hospital environments, data collection and preprocessing can be distributed across multiple processing nodes.

Third, the machine learning component can be retrained with new datasets and emerging attack scenarios, allowing the system to adapt as IoMT networks evolve.

These characteristics make the architecture suitable for both small clinical environments and larger hospital infrastructures.

4.7 Security Considerations

Since the proposed solution operates in a sensitive healthcare environment, security must be considered throughout the architecture.

The following measures are important:

- **Data confidentiality** must be maintained when handling network logs and healthcare-related communications.
- **Access control mechanisms** should restrict unauthorized access to datasets, prediction outputs, and system dashboards.

- **Data anonymization** should be applied where necessary to ensure that no patient-identifiable information is exposed.
- **Secure storage** must be used for historical logs, extracted features, and trained models.
- **Model integrity protection** is necessary to prevent tampering with trained forecasting models.
- **Encrypted communication** should be used if the architecture is extended into real hospital deployments.

These security measures ensure that the proposed system supports healthcare cybersecurity without introducing additional privacy or operational risks.

5. USER REQUIREMENTS

This section defines the user requirements for the proposed **Predictive IoMT Threat Forecasting Engine**. These requirements identify the key stakeholders who interact with the system and outline the functional and non-functional capabilities required to ensure the system operates effectively within healthcare cybersecurity environments.

5.1 Stakeholder Identification

The proposed system will involve several stakeholders who either directly interact with the system or benefit from its outputs. These stakeholders are responsible for managing, monitoring, and maintaining the cybersecurity posture of healthcare IoMT environments.

1. Hospital IT Security Administrators

Hospital cybersecurity teams are the primary users of the system. They are responsible for monitoring IoMT network activity, identifying potential cyber threats, and implementing defensive measures. The predictive forecasting engine assists them by providing early threat warnings and risk analysis.

2. Healthcare IT Infrastructure Managers

These stakeholders manage hospital networks, medical device connectivity, and infrastructure security. They rely on the system to understand network behavior trends and to ensure the stability and safety of IoMT devices within the healthcare environment.

3. Cybersecurity Analysts

Cybersecurity analysts analyze threat intelligence and investigate suspicious activity within hospital networks. The system provides them with predictive insights and threat patterns that support proactive threat mitigation strategies.

4. Medical Device Management Teams

These stakeholders are responsible for maintaining and monitoring connected medical devices such as imaging systems, infusion pumps, and wearable monitoring devices. Predictive threat forecasts help them detect potential risks affecting these devices.

5. Research and Academic Supervisors

Since this system is part of an academic research project, supervisors and researchers will also interact with the system during evaluation and experimentation phases to assess model performance and system effectiveness.

5.2 Functional Requirements

The following functional requirements describe the core operations that the proposed system must perform.

FR1: Data Ingestion

The system shall collect IoMT network traffic data and security logs from selected datasets or healthcare simulation environments.

FR2: Data Preprocessing

The system shall preprocess collected data by performing cleaning, normalization, and feature extraction to prepare it for machine learning analysis.

FR3: Feature Engineering

The system shall identify and extract relevant network features such as traffic patterns, communication behavior, and temporal characteristics required for predictive modeling.

FR4: Threat Pattern Analysis

The system shall analyze historical IoMT network data to identify suspicious activity patterns and anomalies related to cyber threats.

FR5: Predictive Threat Forecasting

The system shall apply machine learning algorithms to analyze historical data and forecast potential cyber threats in IoMT environments.

FR6: Threat Risk Scoring

The system shall generate threat risk levels (e.g., low, medium, high) based on the predicted probability of future cyberattacks.

FR7: Visualization and Reporting

The system shall present forecasted threats through visual dashboards, alerts, and analytical reports for security administrators.

FR8: Model Training and Updating

The system shall allow machine learning models to be retrained with updated datasets to improve prediction accuracy and adapt to evolving cyber threats.

FR9: Threat Alert Generation

The system shall generate warning alerts when predicted threat levels exceed predefined risk thresholds.

FR10: Data Storage

The system shall store processed datasets, extracted features, prediction outputs, and trained models for future analysis and system improvements.

5.3 Non-Functional Requirements

Non-functional requirements define the system's operational quality attributes, including security, performance, reliability, and usability.

Security Requirements

The system must ensure that all collected network data is securely stored and protected from unauthorized access. Access to system dashboards and datasets must be restricted through authentication and authorization mechanisms. Additionally, sensitive healthcare-related data must be anonymized when necessary.

Performance Requirements

The system should efficiently process large volumes of IoMT network data and generate predictions within a reasonable time frame. The machine learning models should be optimized to handle large datasets without significant performance degradation.

Scalability Requirements

The architecture should support scalability so that the system can handle increasing numbers of IoMT devices and larger healthcare network environments in the future.

Reliability Requirements

The system must consistently provide accurate threat predictions and maintain stable operation during data processing and model execution.

Usability Requirements

The system interface should present forecast results in a clear and understandable format. Visualization dashboards and reports should allow administrators to easily interpret predicted threat levels and make informed security decisions.

Maintainability Requirements

The system design should allow easy modification, updates, and integration of new machine learning models or datasets as cybersecurity threats evolve.

Compatibility Requirements

The system should be compatible with commonly used data science platforms and cybersecurity tools to allow integration with existing research and monitoring environments.

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The target market of the proposed system includes organizations whose environments are infused with Internet of Medical Things (IoMT) technologies. This includes hospitals and healthcare providers whose environments are infused with medical devices, medical imaging technologies, and clinical information systems. However, environments infused with such technologies are highly vulnerable to sophisticated cyber attacks such as fileless malware attacks and lateral movement attacks.

The target market includes

- Hospitals and Healthcare Networks
- Medical Research Institutes
- Diagnostic Laboratories
- Telemedicine Service Providers
- Healthcare Cybersecurity Service Providers

The customer personas of the proposed system are

- Hospital IT Security Administrators

These are administrators responsible for monitoring hospital environments. They are responsible for the security of medical devices and patient information. This persona needs a system capable of detecting abnormal device behaviors and cyber-attacks.

- Healthcare Cybersecurity Analysts

These are security analysts working in hospital security operations centers (SOC) responsible for analyzing security alerts to identify potential cyber-attacks.

- Medical Device Security Teams

These are teams responsible for the security of medical devices such as those infused with IoMT technologies.

6.2 Value Proposition

The proposed system provides a hybrid AI-based cybersecurity solution, particularly designed for IoMT in healthcare environments, overcoming the shortcomings of existing security monitoring systems.

The main value propositions are,

- Multi-layer threat detection: The system can detect threats on multiple levels, including network, endpoint, predictive threat patterns, and cross-layer attack correlation.

- Detection of stealthy fileless malware attacks: The system will be particularly designed for detection of memory-based malware attacks, which are difficult for traditional antivirus software to detect.

- Cross-layer attack correlation: The system will be able to correlate attacks from different components, thus providing a comprehensive attack chain

- Explainable AI-based threat interpretation: The system will be able to explain threats detected, helping security analysts understand the rationale for the detection.

- Healthcare-specific cybersecurity protection: Unlike existing intrusion detection systems, the proposed system will be particularly designed for IoMT in healthcare environments, medical imaging, and hospital infrastructures.

6.3 Revenue Model

The proposed system has the opportunity for revenue generation based on the Software-as-a Service (SaaS) or enterprise security solutions business model.

The possible revenue models for the proposed system are:

1. Subscription-Based Model

The proposed system has the opportunity for generating revenue by charging a subscription fee from the healthcare organizations that utilize the proposed system for their operations.

Example:
The proposed system has the opportunity for generating revenue by charging a subscription fee from the healthcare organizations that utilize the proposed system for their operations.
The proposed system has the opportunity for generating revenue by charging a subscription fee from the healthcare organizations that utilize the proposed system for their operations.

2. Enterprise Licensing Model

The proposed system has the opportunity for generating revenue by allowing hospitals to purchase the proposed system for implementation in their hospital's internal cybersecurity infrastructure.

3. Security Monitoring Services

The proposed system has the opportunity for generating revenue by allowing security teams to utilize the proposed system for monitoring security threats.

4. Integration with Existing Security Platforms

The proposed system has the opportunity for generating revenue by allowing the integration of the proposed system into SIEM/SOC systems, which are utilized by cybersecurity vendors for their operations.

6.4 Pricing Strategy

The pricing strategy will be based on the tiered pricing model according to the complexity of the healthcare infrastructure.

Package	Targeted organization	Estimated Price
Basic	Small Clinic centers and laboratories	LKR.150,000.
Professional	Medium Size hospitals	LKR.450,000.
Enterprise	Large hospital networks	Customized price.

Additional services like advanced threat analysis, vulnerability assessment, security consulting services may be provided as premium features.

6.5 Competitive Advantage

The proposed system has several competitive advantages over existing cybersecurity solutions.

- Healthcare-specific security design

The cybersecurity solutions available in the market are generic and can be applied to any general enterprise network. The proposed system is specifically designed for an IoMT healthcare network.

- Hybrid multi-component architecture

The system comprises four different detection components. The combination of these components can provide a comprehensive detection capability for the different layers.

- Cross-layer attack correlation capability

Unlike existing intrusion detection systems, the proposed system can correlate the detected attacks across different layers.

- Explainable AI integration

The inclusion of an XAI dashboard can help in decision-making for cybersecurity analysts.

- Proactive threat forecasting

The inclusion of a predictive threat forecasting engine can help organizations forecast potential threats.

The proposed system can offer a comprehensive cybersecurity solution compared to an existing intrusion detection system or SIEM.

6.6 Intellectual Property Considerations

It is possible that the system may incorporate intellectual properties that can be protected under different intellectual property protection schemes. Possible intellectual properties that may be protected include:

- Software Copyright Protection

It is possible that the software platform, as well as the algorithms for correlation, AI, and dashboard systems, may be protected under software copyright protection.

- 2. Patent Protection Opportunities

It is possible that certain technical aspects of the system may be patented, such as: o Cross-layer correlation algorithms for IoMT cybersecurity. o AI-based attack path reconstruction mechanisms. o Explainable AI dashboards for healthcare cybersecurity.

- 3. Trade Secrets Protection

It is possible that certain aspects of the system, such as:

o Proprietary detection algorithms.

o Threat correlation logic.

o Predictive threat forecasting models

may be protected as trade secrets for commercialization purposes.

It is also necessary to ensure compliance with healthcare data protection regulations such as HIPAA-like regulations and national data protection policies when deploying the system in a real-world environment

6. WORK BREAKDOWN STRUCTURE (WBS)

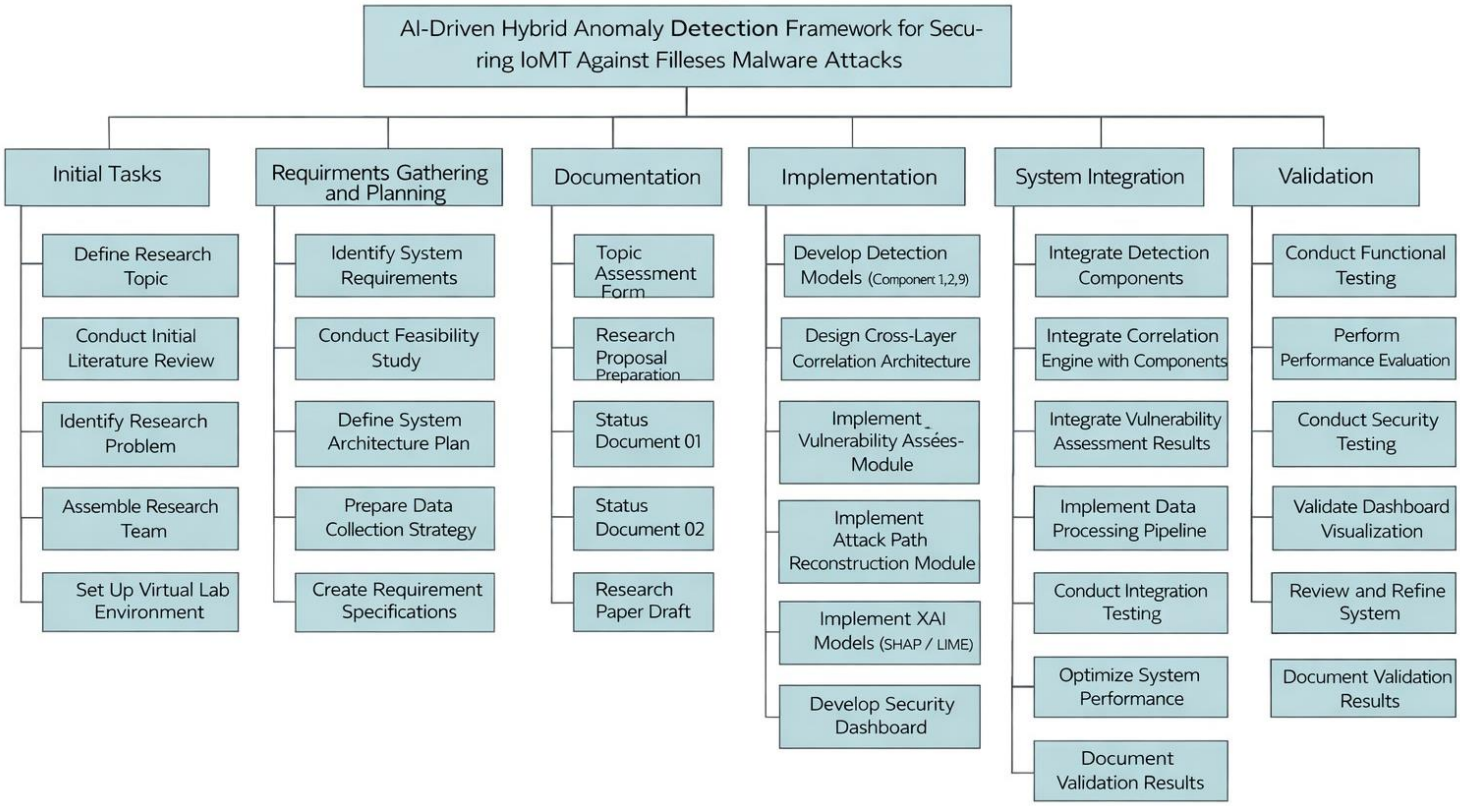


Figure 6.1: Work Breakdown Structure

7. GANTT CHART

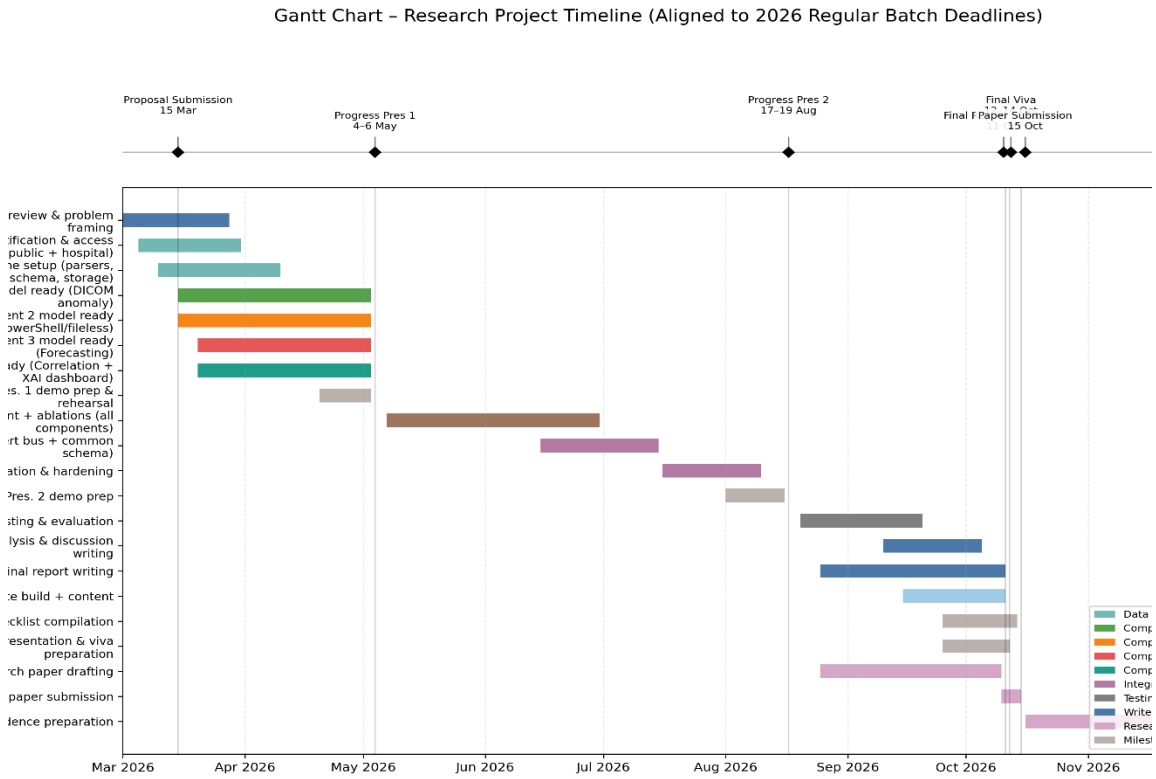


Figure 7.2: Overall research project gantt chart

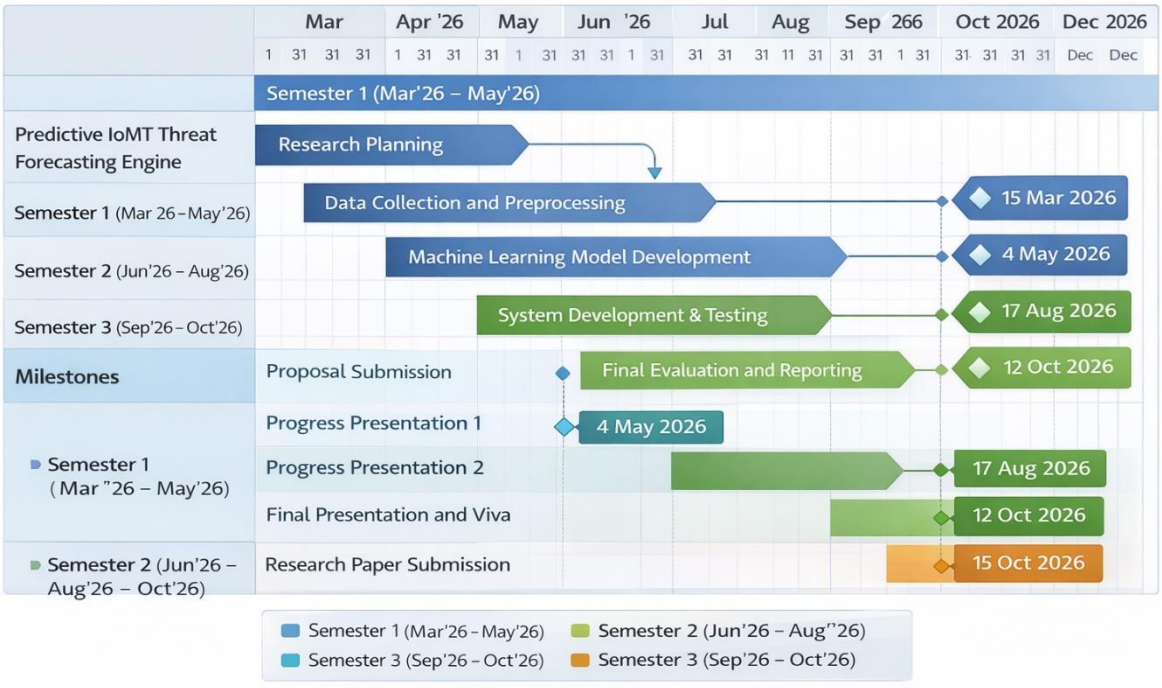


Figure 7.1: Component 3 gantt chart.

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